

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Computer Science and Engineering)(CS201)
Master Subject Coordinator Name:	Ankita Sharma	Master Subject Coordinator E-Code:	E11389
Course Name	COMPUTATIONAL INTELLIGENCE AND PATTERN RECOGNITION AND LEARNING	Course Code	22CSH-463

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
2	0	2	0		0	0	0	0	3.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Elective	Graded (GR)	Hybrid	Hybrid (HYB)

Mission of the Department	M1: To provide practical knowledge using state-of-the-art technological support for the experiential learning of our students. M2: To provide industry recommended curriculum and transparent assessment for quality learning experiences. M3: To create global linkages for interdisciplinary collaborative learning and research. M4: To nurture advanced learning platform for research and innovation for students' profound future growth. M5: To inculcate leadership qualities and strong ethical values through value based education.
Vision of the Department	To be recognized as a leading Computer Science and Engineering department through effective teaching practices and excellence in research and innovation for creating competent professionals with ethics, values and entrepreneurial attitude to deliver service to society and to meet the current industry standards at the global level.

Program Educational Objectives(PEOs)

PEO1	Engage in successful careers in industry, academia, and public service, by applying the acquired knowledge of Science, Mathematics and Engineering, providing technical leadership for their business, profession and community.
PEO2	Establish themselves as entrepreneur, work in research and development organization and pursue higher education.
PEO3	Exhibit commitment and engage in lifelong learning for enhancing their professional and personal capabilities.

Program Specific OutComes(PSOs)

PSO1	Exhibit attitude for continuous learning and deliver efficient solutions for emerging challenges in the computation domain.
PSO2	Apply standard software engineering principles to develop viable solutions for Information Technology Enabled Services (ITES).

Program OutComes(POs)

PO1	Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals and computer science fundamental and strategies which have the solution of complex computer science engineering problems.
PO2	Problem analysis: Identify, formulate, research literature, and analyze complex computer science engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex database and software engineering problems and design system components or processes that meet the specified needs with appropriate considerations for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of software engineering & networking based experiments, analysis and Interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern Computer science engineering and IT tools including prediction and modeling to complex database or software engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess Social, health, safety, legal and cultural issues and the consequent responsibilities relevant to the Professional Computer Science & Engineering practice.

PO7	Environment and sustainability: Understand the impact of the professional computer science and engineering solutions in social and environmental contexts, and demonstrate the knowledge of, and need for sustainable development goals.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of computer science engineering practice
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex computer science engineering activities with the engineering community like CSI society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the computer science engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context to technological change.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Pattern Recognition and Machine Learning	Christopher?M. Bishop	1st Edition (Information Science and Statistics series)	Springe	2006
2	Pattern Recognition and Computational Intelligence Techniques Using MATLAB	E.?S. Gopi	1st Edition	Springer cham	2020

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Computational Intelligence for Pattern Recognition	Witold?Pedrycz, Shyi-Ming Chen	1st Edition	Springer Cham	2018
2	Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence	Jyh-Shing Roger Jang	1st Edition	Pearson Education	2010

Course OutCome	
SrNo	OutCome
CO1	Summarise and depict soft computing methods and their roles to build intelligent terms
CO2	Illustrate fuzzy principles and thinking to deal with vulnerability and tackle real-time issues.
CO3	Apply performance evaluation methods for pattern recognition, and critique comparisons of techniques made in the research literature.
CO4	Demonstrate the performance evaluation methods for pattern recognition, and critique comparisons of techniques made in the research literature.
CO5	Implement simple pattern classifiers, classifier combinations, and structural pattern recognizers.

Lecture Plan Preview-Theory							
Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Number(s)	BT Level
1	1	Foundations of Computational Intelligence	Artificial Intelligence: A Brief Review (definition, evolution, AI vs CI)	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT2

1	2	Foundations of Computational Intelligence	Pitfalls of Traditional Artificial Intelligence	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT1
1	3	Foundations of Computational Intelligence	Need of Computational Intelligence and Its Applications	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT1
1	4	Foundations of Computational Intelligence	Computational Intelligence Techniques (Fuzzy Logic, Neural Networks) & Importance of Tolerance of Imprecision and Uncertainty	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT2
1	5	Introduction to Fuzzy Logic	Constituent Techniques of Fuzzy Logic & Evolutionary Computation	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT2
1	6	Introduction to Fuzzy Logic	Representative Algorithms of Evolutionary Computation	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT1
1	7	Introduction to Fuzzy Logic	Overview of Artificial Neural Networks, Hybrid Approaches & Swarm Intelligence Algorithms	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT2
1	8	Fuzzy Systems and Evolutionary/ Swarm-Based Computational Methods	Swarm-Based Computational Methods Fuzzy Sets and Properties of Fuzzy Sets	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT1
1	9	Fuzzy Systems and Evolutionary/ Swarm-Based Computational Methods	Swarm-Based Computational Methods Membership Functions, Fuzzy Operations & Fuzzy Inference Systems	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT2
1	10	Fuzzy Systems and Evolutionary/ Swarm-Based Computational Methods	Swarm-Based Computational Methods Applications of Fuzzy Logic, Agent-Based Cooperative Optimization & Other Bio-Inspired Algorithms	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT1
2	11	Optimization Fundamentals	Optimization Fundamentals & Descent Methods	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT2
2	12	Optimization Fundamentals	Steepest Descent Method (Derivative-Based Optimization)	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
2	13	Optimization Fundamentals	Classical Newton's Method & Step Size Determination	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT2

2	14	Optimization Fundamentals	Derivative-Free Optimization: Swarm Optimization, Random Search & Downhill Simplex Search	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
2	15	Evolutionary & Metaheuristic Techniques – Introduction	Introduction to Evolutionary Computing & Concepts of Genetic Algorithms	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT2
2	16	Evolutionary & Metaheuristic Techniques – Introduction	GA Techniques & Generational Cycle	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
2	17	Evolutionary & Metaheuristic Techniques – Introduction	Simulated Annealing, Green Computing & Big Data Mining	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO1	BT2
2	18	Pattern Recognition Systems	Introduction, Basic Concepts & Applications of Pattern Recognition	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
2	19	Pattern Recognition Systems	PR System Design: Fundamental Problems, Design Concepts & Methodologies, Simple PR Model	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
2	20	Pattern Recognition Systems	Decision & Distance Functions, Linear & Generalized Decision Functions, Minimum-Distance Classification	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT2
3	21	Clustering and feature extraction techniques	Introduction to Clustering & Feature Selection	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT2
3	22	Clustering and feature extraction techniques	Distance Measures & Clustering Transformation	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT2
3	23	Clustering and feature extraction techniques	Feature Ordering & Entropy-Based Feature Selection	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
3	24	Clustering and feature extraction techniques	Orthogonal Expansion & Binary Feature Selection	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO2	BT2
3	25	Syntactic Pattern Recognition Foundations	Syntactic Pattern Description & Recognition Grammars	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO3	BT3
3	26	Syntactic Pattern Recognition Foundations	Automata as Pattern Recognisers	,T-Pattern Recognition and Machin,R-Computational Intelligence for	PPT,Video Lecture	CO3	BT2

3	27	Syntactic Pattern Recognition Foundations	Structural Pattern Analysis & Syntax-Directed Classification	,T-Pattern Recognition and Machin,R-Computational Intelligence for	PPT,Video Lecture	CO3	BT3
3	28	Applications of Pattern Recognition	Applications of Pattern Recognition Systems	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO4	BT3
3	29	Applications of Pattern Recognition	Biometric Systems: Facial Recognition & IRIS Scan	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT,Video Lecture	CO4	BT3
3	30	Applications of Pattern Recognition	Fingerprint & Voice Recognition Techniques	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO5	BT3

Lecture Plan Preview-Practical

Unit No	ExperimentNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Number(s)	BT Level
1	1	Write a program to analyse Fuzzy logic, Crisp Logic and fuzzy operations.	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	Case Study,PPT	CO1	BT1
1	2	Write a program to implement the Fuzzy Logic System for Input-Output Mapping.	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO2	BT3
1	3	Write a program to develop a Neuro-Fuzzy Model Using ANN Learning to Train/Optimize the Fuzzy System.	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO2	BT2
1	4	Write a program for implementation of Swarm Intelligence Algorithms (PSO / ACO) for Optimization Problems.	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO2	BT2
2	5	Write a program for the implementation of Derivative-Based Optimization Techniques (Steepest Descent and Newton's Method).	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO3	BT2
2	6	Write a program to Implement the Random Search and Downhill Simplex Method.	,T-Pattern Recognition and Comput,T-Pattern Recognition and Machin,R-Computational Intelligence for,R-Neuro-Fuzzy and Soft Computing	PPT	CO2	BT2

3	7	Write a program for implementation of Genetic Algorithms and Simulated Annealing for Function Optimization	,T-Pattern Recognition and Comput,T- Pattern Recognition and Machin,R- Computational Intelligence for,R-Neuro- Fuzzy and Soft Computing	PPT	CO4	BT3
3	8	Write a program for the implementation of feature selection using image Data set.	,T-Pattern Recognition and Comput,T- Pattern Recognition and Machin,R- Computational Intelligence for,R-Neuro- Fuzzy and Soft Computing	PPT	CO4	BT3
3	9	Write a program to apply Pattern Recognition techniques to real world problems such as image analysis, character recognition, etc.	,T-Pattern Recognition and Comput,T- Pattern Recognition and Machin,R- Computational Intelligence for,R-Neuro- Fuzzy and Soft Computing	PPT	CO4	BT3
3	10	Write a program for binary Feature Selection Using Search-Based Optimization Techniques	,T-Pattern Recognition and Comput,T- Pattern Recognition and Machin,R- Computational Intelligence for,R-Neuro- Fuzzy and Soft Computing	PPT	CO5	BT3

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	Practical Evaluations	40	20
2	End Term Hybrid Theory	60	30
3	Attendance Marks	2	2
4	Surprise Test	12	4
5	Practical MST	10	4
6	Practical Worksheet/Projects 1	30	2
7	Practical Worksheet/Projects 2	30	2
8	Practical Worksheet/Projects 3	30	2
9	Practical Worksheet/Projects 4	30	2
10	Practical Worksheet/Projects 5	30	2
11	Practical Worksheet/Projects 6	30	2
12	Practical Worksheet/Projects 7	30	2
13	Practical Worksheet/Projects 8	30	2
14	Practical Worksheet/Projects 9	30	2

15	Practical Worksheet/Projects 10	30	2
16	Quiz	4	4
17	Assignment/PBL	10	6
18	MST-1 Hybrid	20	5
19	MST-2 Hybrid	20	5

